

MODULE DESCRIPTION FORM

نموذج وصف المادة الدراسية

Module Information				
معلومات المادة الدراسية				
Module Title	Advanced Programming Language		Module Delivery	
Module Type	Core		☒ Theory ☒ Lecture ☒ Lab ☒ Tutorial ☒ Practical ☐ Seminar	
Module Code	COMP102			
ECTS Credits	7			
SWL (hr/sem)	175			
Module Level	1	Semester of Delivery		1
Administering Department	CS	College	CS and Mathematics	
Module Leader	Dr. Ali Abdulkarem Habib		e-mail	alia.alramahi@uokufa.edu.iq
Module Leader's Acad. Title	Assistant Professor		Module Leader's Qualification	Ph.D.
Module Tutor	Dr. Ali Abdulkarem Habib		e-mail	alia.alramahi@uokufa.edu.iq
Peer Reviewer Name	Name	e-mail	E-mail	
Scientific Committee Approval Date	01/06/2023	Version Number	1.0	

Relation with other Modules			
العلاقة مع المواد الدراسية الأخرى			
Prerequisite module	Programming Fundamentals	Semester	1
Co-requisites module	Object-Oriented Programming	Semester	3
Co-requisites module	Data Structure	Semester	3
Co-requisites module	Website Design	Semester	5

Module Aims, Learning Outcomes and Indicative Contents

أهداف المادة الدراسية ونتائج التعلم والمحتويات الإرشادية

Module Objectives أهداف المادة الدراسية	This module aims • Optimal use of one dimensional arrays to store data and control indexes. • Optimal use of two dimensional arrays to store data and control indexes. • Teaching students how to use java method and call it whatever we need in the right position in the program • To study different types of java methods
Module Learning Outcomes مخرجات التعلم للمادة الدراسية	1. Knowledge and understanding 2. Understanding the syntax for declaring and allocating one- and two-dimensional arrays, in addition to creating a user-defined methods and knowing how to call it. 3. Cognitive skills (thinking and analysis). 4. After the understanding the syntax of programming tools , the student will have the main requirements for analysis problems and solve them using java code. 5. Communication skills (personal and academic). 6. This course learns the student to be work within a group. 7. Practical and subject specific skills (Transferable Skills). 8. Learn the main function in java language for advance programming tools like array and methods.
Indicative Contents المحتويات الإرشادية	This course will provide an new tools for java programming. How to deal with arrays is the main goal of this course. Using methods in different ways and for different uses is also one of the goals of the course. Learning how to collect data of one type with one array and how to reduce code size using methods is another goal. Attempting to merging the programming tools of the previous course and the current tools to remind them constantly is the last aim of the course.

Learning and Teaching Strategies

استراتيجيات التعلم والتعليم

Strategies	The main strategy that will be adopted in delivering this module is to encourage students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials, practical, lab hours and by considering types of some small projects that are interesting to the students.
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Student Workload (SWL)			
الحمل الدراسي للطلاب محسوب ل ١٥ اسبوعا			
Structured SWL (h/sem) الحمل الدراسي المنتظم للطلاب خلال الفصل	110	Structured SWL (h/w) الحمل الدراسي المنتظم للطلاب أسبوعيا	7
Unstructured SWL (h/sem) الحمل الدراسي غير المنتظم للطلاب خلال الفصل	65	Unstructured SWL (h/w) الحمل الدراسي غير المنتظم للطلاب أسبوعيا	6
Total SWL (h/sem) الحمل الدراسي الكلي للطلاب خلال الفصل	175		

Module Evaluation					
تقييم المادة الدراسية					
		Time/Number	Weight (Marks)	Week Due	Relevant Learning Outcome
Formative assessment	Quizzes	2	10% (10)	5 and 10	All
	Assignments	2	10% (10)	4 and 12	All
	Projects / Lab.	1	10% (10)	Continuous	All
	Report	1	10% (10)	13	All
Summative assessment	Midterm Exam	2hr	10% (10)	7	All
	Final Exam	3hr	50% (50)	16	All
Total assessment			100% (100 Marks)		

Delivery Plan (Weekly Syllabus)	
المنهاج الاسبوعي النظري	
	Material Covered
Week 1+2	Java String : Types of java methods on string with examples
Week 3+4	One Dimensional Array with examples on one dimensional arrays
Week 5+6	Two dimensional Array with examples on two dimensional arrays
Week 7	First Exam
Week 8+9	Java Methods: Types of methods
Week 10-11,12	Examples on Java Methods
Week 13	Second Exam
Week 14	Revision and project discussion
Week 15	Final Exam

Delivery Plan (Weekly Lab. Syllabus) المناهج الاسبوعي للمختبر	
	Material Covered
Week 1	Lab 1: Write program to apply all methods of string
Week 2	Lab 2: Simple java program on static method
Week 3	Lab 3: writes programs on 1 D array
Week 4	Lab 4: solve assignments programs on 1 D array
Week 5	Lab 5: writes programs on 2 D array
Week 6	Lab 6: solve assignments programs on 2 D array
Week 7	Lab 7: write all examples on 1D and 2D arrays
Week 8	First Exam
Week 9	Lab 8: write programs on method with no return value and no pass parameter
Week 10	Lab 9: write programs on method with no return value and pass parameter
Week 11	Lab 10: write programs on method with return value and pass parameter
Week 12	Lab 11: Second Exam
Week 13	Lab 12: Write java programs in assignments
Week 14	Lab 12: Discussion assignments
Week15	Final Exam

Learning and Teaching Resources مصادر التعلم والتدريس		
	Text	Available in the Library?
Required Texts	Java How to Program (9 th Edition) ,Paul and Harvey Deitel,2011.	Yes
Recommended Texts	Java for Absolute Beginners , Iuliana Cosmina, 2018	Yes
Websites	• https://www.w3schools.com/java/ • https://www.codecademy.com/learn/learn-java • https://www.javatpoint.com/java-tutorial	

Grading Scheme مخطط الدرجات				
Group	Grade	التقدير	Marks %	Definition
Success Group (50 - 100)	A - Excellent	امتياز	90 - 100	Outstanding Performance
	B - Very Good	جيد جدا	80 - 89	Above average with some errors
	C - Good	جيد	70 - 79	Sound work with notable errors
	D - Satisfactory	متوسط	60 - 69	Fair but with major shortcomings

	E - Sufficient	مقبول	50 - 59	Work meets minimum criteria
Fail Group (0 – 49)	FX – Fail	راسب (قيد المعالجة)	(45-49)	More work required but credit awarded
	F – Fail	راسب	(0-44)	Considerable amount of work required

Note: Marks Decimal places above or below 0.5 will be rounded to the higher or lower full mark (for example a mark of 54.5 will be rounded to 55, whereas a mark of 54.4 will be rounded to 54. The University has a policy NOT to condone "near-pass fails" so the only adjustment to marks awarded by the original marker(s) will be the automatic rounding outlined above.